

CONTEMPORARY REVIEW

Building Digital Twins for Cardiovascular Health: From Principles to Clinical Impact

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ABSTRACT: The past several decades have seen rapid advances in diagnosis and treatment of cardiovascular diseases and stroke, enabled by technological breakthroughs in imaging, genomics, and physiological monitoring, coupled with therapeutic interventions. We now face the challenge of how to (1) rapidly process large, complex multimodal and multiscale medical measurements; (2) map all available data streams to the trajectories of disease states over the patient's lifetime; and (3) apply this information for optimal clinical interventions and outcomes. Here we review new advances that may address these challenges using digital twin technology to fulfill the promise of personalized cardiovascular medical practice. Rooted in engineering mechanics and manufacturing, the digital twin is a virtual representation engineered to model and simulate its physical counterpart. Recent breakthroughs in scientific computation, artificial intelligence, and sensor technology have enabled rapid bidirectional interactions between the virtual-physical counterparts with measurements of the physical twin that inform and improve its virtual twin, which in turn provide updated virtual projections of disease trajectories and anticipated clinical outcomes. Verification, validation, and uncertainty quantification builds confidence and trust by clinicians and patients in the digital twin and establishes boundaries for the use of simulations in cardiovascular medicine. Mechanistic physiological models form the fundamental building blocks of the personalized digital twin that continuously forecast optimal management of cardiovascular health using individualized data streams. We present exemplars from the existing body of literature pertaining to mechanistic model development for cardiovascular dynamics and summarize existing technical challenges and opportunities pertaining to the foundation of a digital twin.

Key Words: cardiovascular modeling ■ computational physiology ■ digital representation ■ precision health

A digital twin for precision health is a set of virtual information constructs that mimics the structure, context, and behavior of a human body or health systems (or system-of-systems). It is dynamically and continuously updated with data from its physical twin. Thus, it has a predictive capability. Its correctness can be verified and informs decisions that realize value and provide actionable information to guide health and

wellness care delivery. The bidirectional flow of information from the human health system to the computational model affects a twin that remains tightly coupled with the human health system leading to more effective identification of risk factors in the presence of current or future behavior or adverse events.^{1,2} This information flow in continuous cycles is enabled by sensors continually providing data to update the digital twin, which

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Nonstandard Abbreviations and Acronyms

CSA	Coordination and Support Action
EDITH	Ecosystem Digital Twins in Health
FDA	Food and Drug Administration
PINN	Physics-Informed Neural Network

then provides feedback about the physical twin's current state and possible future states in the form of calculated predictions.^{3,4} Digital twin representations have been used successfully in the fields of aerospace and construction, which have provided insights for airplanes and buildings to be designed and simulated digitally for safety and functionality.^{5,6} The digital twin concept has also been proposed in multiple areas of health care, including cardiology,^{7–9} oncology,¹⁰ and diabetes treatment,¹¹ with a focus on precision health modeling. Although these early versions of a digital twin successfully demonstrated potential in precision medicine, they exhibit a limited scope, concentrating on the complexity of individual disease models. A digital twin should evolve beyond singular disease modeling and instead strive toward comprehensive modeling of physiological systems, necessitating continuous updates for the digital twin to adapt providing a foundation for robust predictive capabilities.

It is worth noting that various definitions of digital twins have been introduced in the past, describing the mechanisms pertaining to digital twin construction, operation and expected output. One definition that has garnered considerable attention is by the EDITH (Ecosystem Digital Twins in Health) Coordination and Support Action funded by the European Commission.¹² EDITH defines digital twin for health as computer simulations, incorporating both knowledge-driven and data-driven models to predict clinically relevant quantities otherwise challenging to experimentally measure. These digital twins should also enable forecasting of the evolution of such quantities as well as their response to external actions. They can be constructed as generic, population-specific, or subject-specific (ie, personalized) models based on the degree of accuracy required to meet its purpose. Given the high personalization requirements for many cardiovascular health applications under precision health, our team adopted the definition introduced by the National Academies of Sciences, Engineering, and Medicine,¹ that presents a comprehensive digital twin ecosystem through 5 key elements: (1) creating a personalized virtual representation through modeling and simulations of the physical twin, (2) acquiring observations from the physical twin through sensors and past records, (3) flow from physical to virtual twin to calibrate and update the

digital representation and estimate parameters that are not directly observable, (4) flow from the virtual twin to physical twin to drive positive changes in the physical twin that is initiated by (5) the human in the loop. In the context of cardiovascular health, the digital representation would be an individual's heart and vascular system, where clinical data, such as magnetic resonance imaging (MRI) and medical records, genetic/omic (eg, family history, DNA sequence), demographic (eg, sex, age), and continuous (eg, physiological, biochemical, environmental) data collected with multiple sensor modalities (eg, wearables) would provide the digital twin with the necessary information to construct and continually update itself.¹³ Such measurements, when integrated into the cardiovascular digital twin, will enable the digital twin to accurately and continuously track and mimic the dynamic behavior of the organ system, assess its health status, and provide disease onset predictions well in advance beyond what is currently possible without the exploitation of a digital twin by medical practitioners to instigate more accurate, timely, and informed clinical decisions.^{9,13} The framework for a cardiovascular digital twin is shown in [Figure 1](#).

Cardiovascular digital twins will be able to intimately monitor health at the individual level, enabling continuous feedback on the possibility of an adverse event such as a myocardial infarction, heart failure, or stroke.^{14,15} In addition, they will also be able to provide historical trends of an individual's cardiovascular health by retaining and analyzing important physiological information collected over the span of months and years.¹⁶ This is especially useful for physicians when assessing a patient's health status in reference to their baseline, allowing for improved accuracy and effectiveness in disease diagnosis and personalized treatment planning.⁹ In this way, a cardiovascular digital twin will inform and enhance personalized risk assessment, prevention, and treatment of disease. A cardiovascular digital twin will also enable medical practitioners to simulate outcomes, obtain corresponding health trajectories, and determine the most effective health delivery decisions in view of their patients' unique cardiovascular characteristics, as shown in [Figure 2](#).

Current state-of-the-art cardiovascular system modeling allows us to construct sophisticated mechanistic models of the human heart and vasculature. Such mechanistic models are built upon the foundation of known mechanical, electrical, and statistical theories and leverage biophysical and biomechanical underpinnings.^{17–19} These mechanistic models typically represent an average individual in a large population (using the typical parameter values derived from the data collected from a population of interest) rather than tailored to a specific individual (using the personalized parameter values derived from the data collected from an individual).²⁰ A digital twin must

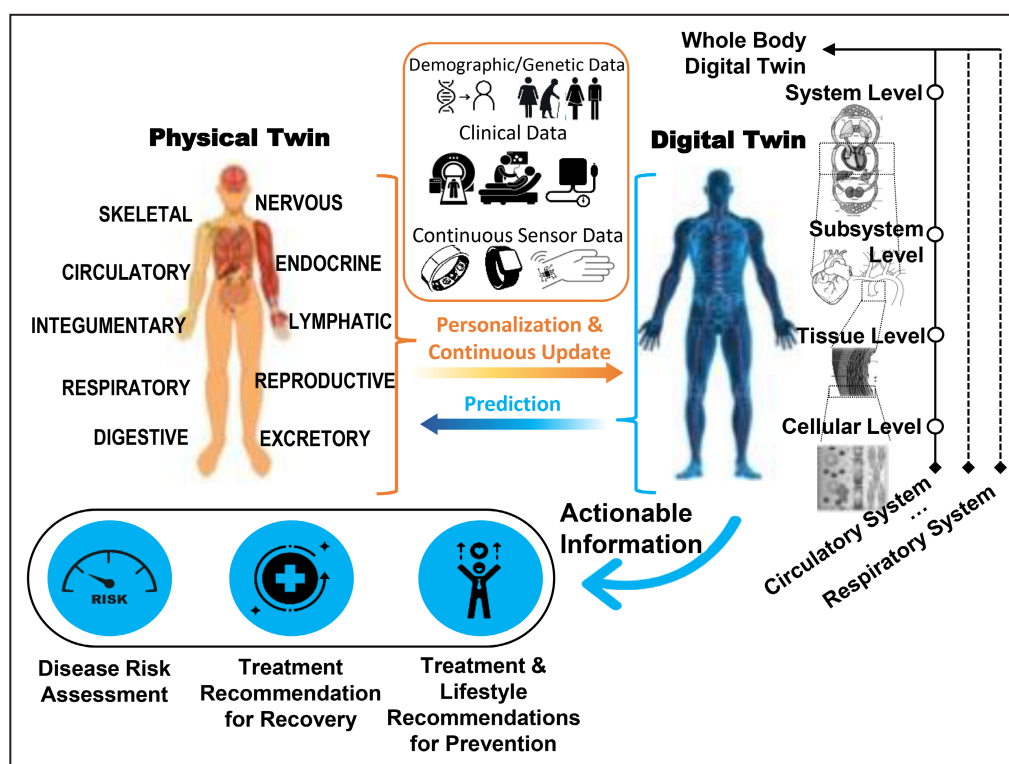


Figure 1. The whole-body digital twin concept incorporating multimodal data sources, multisystem interactions, and hierarchical organ system structures to provide actionable information.

persistently represent the individual using a mathematical model. The difference between a traditional modeling approach driven by mechanistic modeling and a digital twin lies in the ability for seamless learning and update. One opportunity is to leverage machine learning and artificial intelligence (AI) to process continuous, complex, and heterogeneous data obtained from a constantly changing and evolving physical twin to update the state of a mechanistic model.²¹ As a next-generation advancement of these mechanistic models, a digital twin constructed based on the foundation of physiology would offer superior continuous risk assessments by seamlessly integrating data from sensors that acquire vital health metrics. In addition, their interpretability is enhanced due to the mathematical framework inherited from existing mechanistic modeling techniques.²² Another important consideration is in situations where data are limited, misrepresentative, or the problem at hand is exceedingly complex, where mathematical modeling approaches often struggle to provide accurate risk estimations. Here, AI has a potentially large role, as it may provide a surrogate for mechanistic models when they are too complex or fail to represent certain parts of a biological system.²³ AI may also serve as a bridge between mechanistic models and diverse data types, enabling the integration and interpretation of a wide spectrum of information,

from genetic and demographic data to extensive clinical measurements (eg, 4-dimensional functional MRI), as well as real-time physiological data from ambulatory sensors, which would often be incompatible for mechanistic models to process. Moreover, AI's dynamic learning capabilities permit to evolve in tandem with new sensor data, essentially enabling the continuous update of a digital twin so that it can closely reflect an individual's evolving health state.

The development of the cardiovascular digital twin framework incorporating AI with the state-of-the-art mechanistic modeling techniques is meaningful and actionable for medicine only when it represents the patient or individual with sufficient accuracy based on the data obtained in various types, including clinical data, genetic history, and patient characteristics (eg, age, sex), while being continuously updated through sensor data. A digital twin, constructed to represent the body, must mimic the underlying physiological systems and organs. Physiologically, the body represents a hierarchical structure, with cells, tissues, organs, and organ systems. The aspiration for a digital twin is to capture this hierarchical structure and enable hierarchical design and composure of the digital twin with sufficient personalization at multiple levels.⁷ Feasibility of modeling the individual's physiology starting at the molecular scale with its entire depth is still a question,

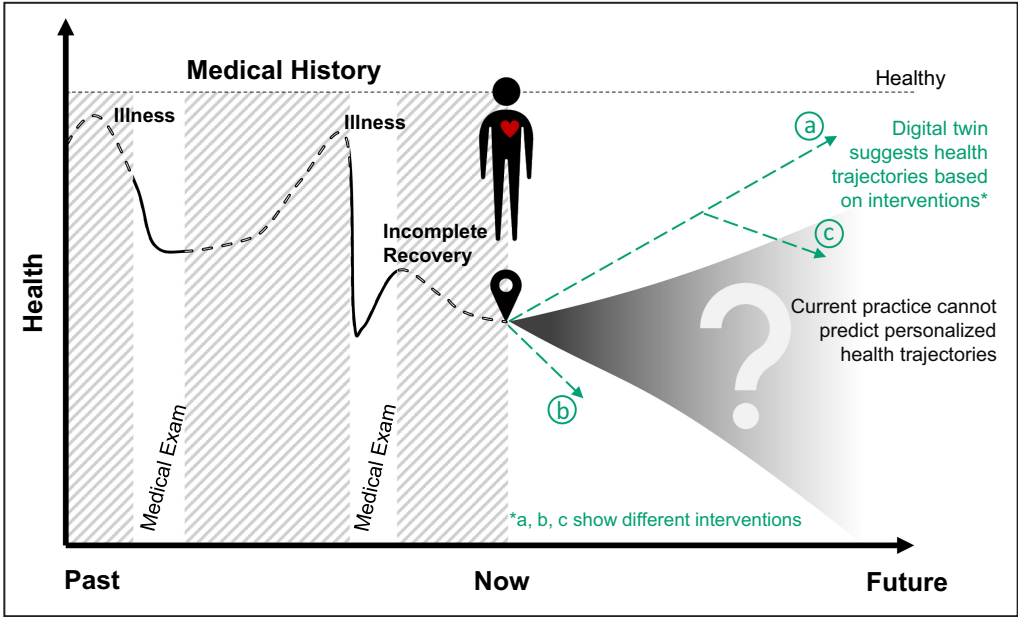


Figure 2. A digital twin can predict personalized health trajectories to allow medical practitioners to determine the most effective personalized intervention paradigms.

due to both theoretical and practical challenges involved. For instance, a “complete” digital twin for an individual starting from the cellular level would require complex quantum biochemical simulations that would overcomplicate the problem in hand with no practical reasoning. Therefore, the level of resolution and personalization for the digital twin should rather be driven by the clinical application with a motivation to provide maximum utility. In any case, the digital twin concept will likely transform the notion of computational medicine beyond what is possible today. The hierarchical

mathematical models embedded in a digital twin for cardiovascular health, constructed with sound theoretical foundation and validated using experiments, can provide a transformative opportunity for personalized health and medicine. This concept of hierarchical design will enable the construction of mathematical models from scratch and in a truly individualized fashion. This direction is important because the current practice of constructing mathematical models often begins at the level of communities, groups, and populations. A mathematical model constructed for groups

Table 1. Overview of Cardiovascular System Modeling Techniques

Subsection title	Data type	Model hierarchy
Computational modeling combining electrophysiology and biomechanics	Clinical data	Molecular, cellular, tissue, organ, and system levels
Cardiovascular modeling for thrombosis	Clinical data	Molecular, cellular, and tissue levels
Cardiovascular models for ballistocardiogram insight	Demographic, clinical, and sensor data	Cellular, tissue, organ, and system levels
Ventricular and atrial tachycardia ablation planning using computational modeling	Demographic and clinical data	Cellular, tissue, and organ levels
Patient-specific ventricular electrophysiological modeling	Clinical and sensor data	Cellular, tissue, and organ levels
Modular multiscale mathematical models for heart failure	Demographic and clinical data	Cellular, tissue, organ, and system levels
3D modeling of an arterial aneurysm	Clinical data	Tissue level
Mechanistic and machine learning modeling for blood pressure and pulse volume waveforms	Demographic and sensor data	Tissue and system levels
Personalized lumped-parameter models for cardiomyopathies	Clinical and sensor data	Tissue and organ level
Interconnected cardiopulmonary modeling	Clinical data	System level
Integrated modeling of cardiorenal system	Clinical data	System level

inherently does not capture the unique characteristics and needs of individuals and will ultimately limit its capabilities in making trustworthy estimations and predictions at the individual level. A mathematical model constructed from the ground up may be fine-tuned and customized to the unique characteristics of individuals. This is a notable direction that will likely create a digital twin that exhibits high degrees of precision beyond what is available today.

A digital twin would ultimately comprise multiple organ systems that continuously interact using data, mimicking the behaviors of physical organ systems as observed in the real world. In addition, the hierarchical architecture of these systems would ideally begin at the cellular level and build up to the system level, ensuring the consistency in the behaviors between the digital twin and its physical counterpart from the microscale to the macroscale level. The significance of this whole-body digital twin is to provide actionable information relevant to personalized disease risk assessment, treatment of disease, and disease prevention. This technological feat would revolutionize medicine as a whole by expediting the paradigm shift toward a continuous personalized approach to health care. By way of example, as highlighted earlier, industry saw significant advancements in the aerospace and construction sectors: computational science precisely realized digital twins of aircrafts and buildings for safety and effectiveness, which in turn supported the accelerated development of engineering innovations in these fields. The bold vision of digital twin for cardiovascular health will likely enable the next important breakthroughs at the intersection of engineering and medicine.

This review work analyzes current cardiovascular system modeling techniques and their ability to monitor the health of the heart and vasculature at individualized levels. [The Review of Cardiovascular System Modeling Techniques](#) section reviews mechanistic models, categorized according to their relevance to preventive cardiovascular medicine and organized according to the modeling hierarchy. [Table 1](#) provides an overview of the presented modeling categories. We highlight the limitations associated with each approach and their integration into a digital twin platform. Our work identifies opportunities for the progression of cardiovascular system modeling based on the integration of mechanistic modeling and AI toward establishing a cardiovascular digital twin and eventually a whole-body digital twin to enhance the continuity and connection of personalized health into the realm of digital medicine. In the [Opportunities for a Cardiovascular Digital Twin](#) section, we highlight future directions and research opportunities inspired by the bold vision of construction of a digital twin for our cardiovascular system, and eventually a full body digital twin incorporating all 11 organ systems.

REVIEW OF CARDIOVASCULAR SYSTEM MODELING TECHNIQUES

Overview

Cardiovascular system modeling plays a crucial role in understanding the complex interactions between the heart, connecting circulation and other organ systems. Furthermore, cardiovascular system modeling has the potential ability to quantitatively study the interactions within the cardiovascular system without the need for invasive procedures. Approaches include mechanistic modeling of these physiological interactions using equations from other scientific fields such as mechanics and electronics. [Figure 3](#) illustrates techniques used to create these cardiovascular system models. [Table 2^{24–52}](#) provides detailed information on the reviewed studies in terms of the implemented modeling techniques, inputs and outputs of the system, modeled components, the level of personalization during the construction of each model, the targeted diseases, and the real-data integration. The studies included in [Table 2](#) are selected from a comprehensive body of recent work on cardiovascular modeling to demonstrate various important modeling approaches and provide valuable insights into how these models addressed specific diseases, incorporated personalized parameters into different anatomical hierarchies (such as cellular, tissue, or organ), and integrated personalized or population-level data. All studies provided in this section focus on specific cardiovascular system models, based upon their novel contributions to the field, significance toward developing a digital twin, and their current limitations.

The reviewed models currently lack key components, hindering their evolution into comprehensive digital twin representations. Two significant shortcomings are the absence of continuous update mechanisms and the lack of personalization for individual physiological parameters. In the subsequent subsections, we outline the parameters (clinical, demographic, etc.) used or potentially employable for personalizing each model. Furthermore, these models are structured according to an anatomical hierarchy. Beginning with models that integrate molecular and cellular modeling techniques, followed by tissue and organ modeling, and concluding with organ system-level modeling. Although these models serve as the foundational basis for a cardiovascular digital twin by incorporating statistical and mechanistic knowledge to describe physiological processes related to the cardiovascular system, existing gaps prevent them from truly embodying the concept of digital twins. To address these limitations, it becomes necessary to focus on integrating continuous monitoring, real-time updates, and personalized parameterization. These enhancements are essential for transforming these models into dynamic digital

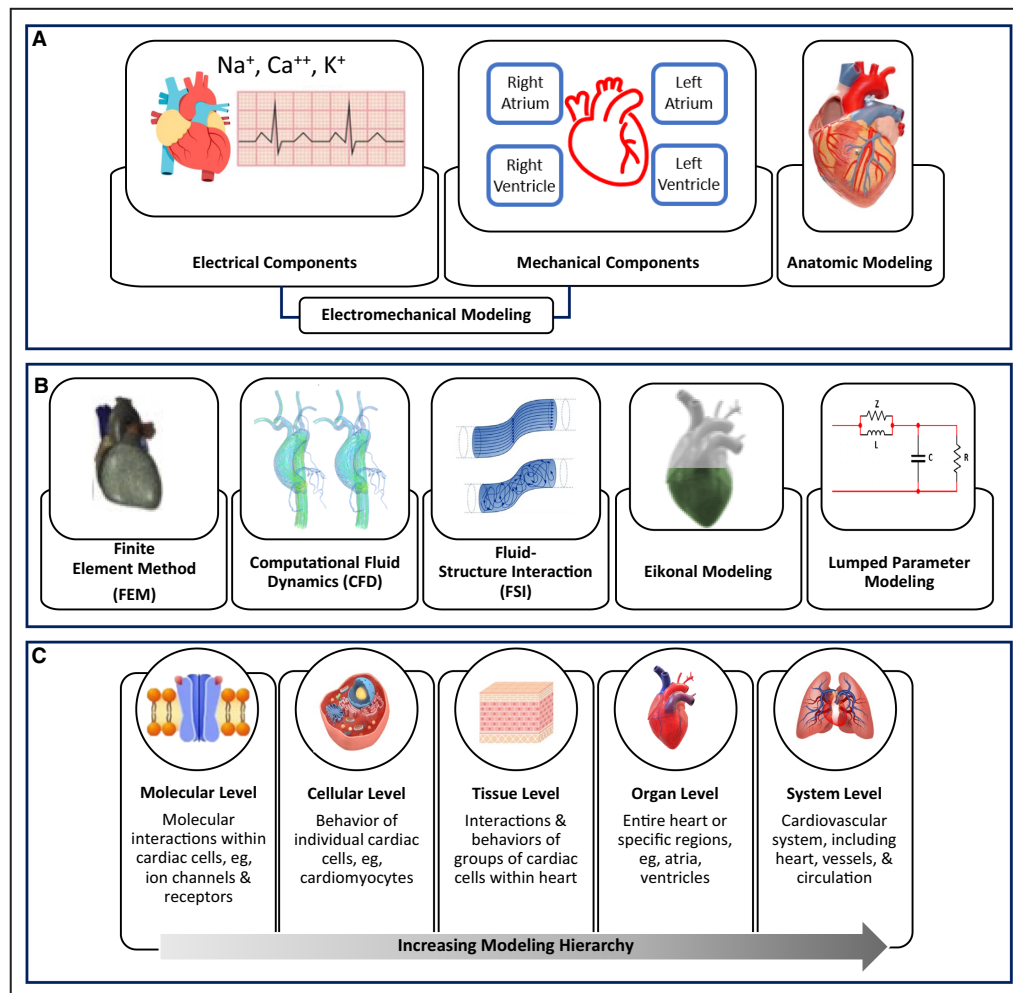


Figure 3. Illustration of technique categories used to create cardiovascular system models. **A.** Physiological aspects of modeling. **B.** Modeling methods. **C.** Increasing modeling hierarchy.

twins that accurately reflect and adapt to the complexities of individualized physiological conditions.

Computational Modeling Combining Electrophysiology and Biomechanics

Cardiac electromechanical models are computational representations of electrophysiological and biomechanical behavior of the heart. These models are used to study and simulate the complex interactions between the electrical signals that drive cardiac contractions and the resulting mechanical response of the myocardium. Electromechanical models offer promising resources to enhance health care, particularly in the context of managing cardiac arrhythmias, such as ventricular tachycardia and fibrillation.⁵³

There has been a lot of effort in developing mechanistic models that can mimic the behavior of the cardiovascular system in the human body. However, there are only a small number of these models that follow the intricate hierarchical structure of the heart, capturing

its biomechanical and electrophysiological characteristics while replicating its overall hemodynamic behavior. These mechanistic models are cohort based and nonpersonalized and far from the idea of a dynamically updating digital twin, but they still have an informative capability that can be used to gain insights about the macroscopic and microscopic behavior of the heart. Five such models are reviewed in the order of their complexity. In one study,³⁷ a mechanistic model of electrophysiology at the level of a human ventricular cell was connected to a mechanistic model of mechanical contraction in a half-sarcomere, which was then incorporated into the sphere-shaped mechanistic models of the ventricles, mapping the pressure and volume of both ventricles to the force and length of a half-sarcomere. In contrast to the common lumped-parameter mechanistic models of the ventricles, this method can explicitly incorporate the physiological mechanisms underlying the cardiac contraction and potentially simulate the hemodynamic changes induced by heart failure solely by altering the cellular properties. In another study,²⁴

Table 2. Detailed Characterization and Categorization of Reviewed Reports on Cardiovascular System Models

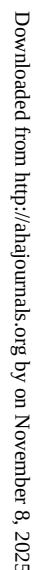
Study (year)	Modeling method	Model input/output	Data type	Modeled component	Level of personalization	Disease	Challenges for DT to address	Real data integration
Watanabe (2004) ²⁴	LPM, FEM, FSI, EMM	N/A hemodynamics & LV P-V curve	N/A	Cardiovascular system	N/A	N/A	No individual personalization, missing data integration, and no continuous update mechanism	N/A
Kerckhoffs (2007) ²⁵	LPM, FEM, and MM	N/A hemodynamics, strain, LV P-V curve	N/A	Cardiovascular system	Disease level	Ischemia and artery constriction	No individual personalization, missing data integration, and no continuous update mechanism	N/A
Bai-Nan (2010) ²⁶	FSI and computational fluid dynamics	CT angiography BP	Clinical	Carotid artery	Patient level	Cerebral aneurysm	No continuous update mechanism	Human subj. N=1
Luo (2011) ²⁷	LPM	N/A respiratory and hemodynamics	N/A	Cardiovascular and respiratory system	Disease level	HFrEF and HFpEF	No individual personalization, missing data integration, and no continuous update mechanism	N/A
Wenk (2013) ²⁸	LPM, FEM, and MM	N/A myofiber stress and LV P-V curve	N/A	Cardiovascular system	Disease level	Dilated cardiomyopathy	No individual personalization, missing data integration, and no continuous update mechanism	Animal model N=1
Krishnamurthy (2013) ²⁹	LPM, FEM, and MM	CT, echocardiography LV P-V curve	Clinical	Cardiovascular system	Disease level	Left ventricular dysfunction	No individual personalization and no continuous update mechanism	Human subj. N=5
Marchesseau (2013) ³⁰	FEM, EMM	MRI LV contractility	Clinical and Demographic	Left ventricle	Disease level	HF	No individual personalization and no continuous update mechanism	Human subj. N=11
Augustin (2015) ³¹	EMM and FEM	Cardiac MRI N/A	Clinical	Heart	Patient level	N/A	No continuous update mechanism	Human subj. N=1
Albanese (2016) ³²	LPM	N/A respiratory and hemodynamics	N/A	Cardiopulmonary system	Disease level	Hypoxic and hypercapnic	No personalization, missing data integration, and no update mechanism	Human subj. N=14
Arevalo (2016) ³³	EPM	Cardiac MRI arrhythmia risk	Clinical	Ventricles	Patient level	Myocardial infarction	No continuous update mechanism	Human subj. N=41
Elizondo (2016) ³⁴	MM	Vein wall shear stress N/A	Clinical	Veins	N/A	Vein thrombosis	No individual personalization and no continuous update mechanism	N/A
Pant (2016) ³⁵	LPM	BP, MRI, Doppler velocimetry hemodynamics	Clinical and sensor	Cardiovascular system	Disease level	Valve regurgitation	No individual personalization	Human subj. N=2
Schiavazzi (2016) ³⁶	LPM	MRI, hemodynamic, Doppler velocity hemodynamic, P-V curves	Clinical and sensor	Cardiovascular system	Diseased level	Valve regurgitation	No individual personalization	Human subj. N=4
Kosta (2017) ³⁷	LPM, EP, and MM	N/A cellular and tissue parameters, LV P-V curve	N/A	Cardiovascular system	Disease level	HF	No individual personalization, missing data integration, and no continuous update mechanism	N/A

(Continued)

Table 2. Continued

Study (year)	Modeling method	Model input/output	Data type	Modeled component	Level of personalization	Disease	Challenges for DT to address	Real data integration
Cedilnik (2018) ³⁸	EP	Cardiac CT ablations	Clinical	Heart	Patient level	Myocardial infarction	No continuous update mechanism	Human subj. N=7
Lee (2018) ³⁹	LPM	Blood volume carotid artery BP	Clinical and sensor	Circulatory system	Patient level	N/A	No continuous update mechanism	Human subj. N=133
Yousefian (2019) ⁴⁰	LPM	BP tissue force	Clinical and sensor	Aorta	Disease level	Cardiovascular disease	No individual personalization	Human subj. N=30
Guidoboni (2019) ⁴¹	LPM	Waveform hemodynamics	Sensor & demographic	Cardiovascular system	Disease level	HFrEF	No individual personalization	Human subj. N=1
Bozkurt (2019) ⁴²	LPM	N/A hemodynamics	N/A	Cardiovascular system	Disease level	Dilated cardiomyopathy	No individual personalization, missing data integration, and no continuous update mechanism	N/A
Yu (2020) ⁴³	LPM	N/A hemodynamics	N/A	Cardiovascular system	Disease level	Diabetic and nondiabetic HFrEF	No individual personalization, missing data integration, and no continuous update mechanism	Human subj. N=108
Shavik (2021) ⁴⁴	LPM and FEM	N/A hemodynamics	N/A	Cardiovascular system	Disease level	HFpEF	No individual personalization, missing data integration, and no continuous update mechanism	Human subj. N=3
Rabineau (2021) ⁴⁵	LPM	Angiography MRI ballistocardiography	Clinical	Cardiovascular system	Phenotype and patient level	N/A	No continuous update mechanism	Human subj. N=1
Gillette (2021) ⁴⁶	LPM	MRI, ECG Ventricular electrophysiology.	Clinical and sensor	Heart	Patient level	N/A	No continuous update mechanism	Human subj. N=12
Van Osta (2021) ⁴⁷	LPM	Echocardiograph myocardial properties	Clinical	Cardiovascular system	Disease level	Arrhythmogenic cardiomyopathy	No individual personalization and no continuous update mechanism	Human subj. N=9
Zaid (2022) ⁴⁸	LPM	N/A ballistocardiography	N/A	Cardiovascular system	Disease level	Myocardial infarction	No individual personalization, missing data integration, and no continuous update mechanism	Animal models N=7
Bozkurt (2022) ⁴⁹	LPM	Hemodynamics, MRI hemodynamics and heart geometry	Sensor and demographic	Cardiovascular system	Disease level	Dilated cardiomyopathy	No individual personalization	Human subj. N=3
Roney (2022) ⁵⁰	EPM	MRI, ECG long-term clinical atrial fibrillation recurrence	Clinical and sensor	Heart	Patient level	Atrial fibrillation	No continuous update mechanism	Human subj. N=100
Fedele (2023) ⁵¹	LPM, FEM, and EMM	N/A P-V curve, cardiac deformation maps	N/A	Cardiovascular system	N/A	N/A	No individual personalization, missing data integration, and no continuous update mechanism	N/A
Azzolin (2023) ⁵²	EPM	MRI, CT areas experiencing atrial fibrillation	Clinical	Heart	Patient level	Atrial fibrillation	No continuous update mechanism	Human subj. N=29

Sorted based on study's publication year. BP indicates blood pressure; CT, computed tomography; EMM, electromechanical model; EPM, electrophysiological model; FEM, finite element method; FSI, fluid–structure interaction; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; LPM, lumped parameter model; LV, left ventricular; MM, mechanical model; MRI, magnetic resonance imaging; and P-V, pressure-volume.



Cardiovascular Modeling for Thrombosis

Veins play a significant role in cardiovascular dynamics and are prone to formation of clots. Mechanistic models simulating the initiation of thrombus formation, using cellular reactions responsible for fibrin formation, have been developed.^{34,55} Models of a thrombin burst using shear stress measured in venous walls and concentrations of multiple coagulation proteins revealed the importance of coagulating proteins such as thrombomodulin and antithrombin.³⁴ In related work, medical imaging and biophysical modeling techniques were employed to model left atrial appendage, which is a primary source for thrombus formation.⁵⁶ A possible improvement to enhance the usability of this mechanistic model for digital twin applications is to incorporate personalization parameters tailoring the mechanistic model for a specific individual and to continuously monitor the concentrations of the specified proteins, which are the patient parameters, contributing to the formation of the venous thrombus. This mechanistic model provides a hierarchical structure for simulating the dynamics of the vein at the cellular level and at the fluid mechanical level for the formation of clots, which is an important concept in the formulation of a comprehensive digital twin. Possible improvements to enhance the usability and feasibility of this mechanistic model for digital twin applications are to incorporate personalization parameters tailoring the mechanistic model for a specific individual and to continuously monitor the concentrations of the specified proteins contributing to the formation of the venous thrombus.

Cardiovascular Models for Ballistocardiogram Insight

Ballistocardiography involves the measurement of periodic movements of the body induced by the heartbeat, specifically by the ejection of blood by heart into aorta. Recent advances in wearable sensors have renewed interest in ballistocardiography as a noninvasive cardiac monitoring and diagnostic technique. However, a comprehensive understanding of the physiological origin of the ballistocardiography signal still needs to be discovered. Compared with other biomedical signals like ECG, following cardiovascular disease effects on ballistocardiography waveforms is challenging. Computational modeling can provide insights into the genesis of ballistocardiography and broaden its clinical utility. Researchers have developed a mechanistic model consisting of a closed-loop 0-dimensional (resistor-capacitor-resistor circuit and resistor-inductor-capacitor circuits) and 1-dimensional (1D; axisymmetric tube) representations of the systemic and pulmonary circulations. After simulating blood flow and pressure throughout the mechanistic model, the ballistocardiogram is generated based on the distribution of the

blood volume and its anatomical location. The mechanistic model is able to reproduce the typical shape and amplitude of the ballistocardiogram reported clinically and studies the impact of aging on the morphology. Furthermore, in modeling healthy aging, modifications are made to parameters such as pulse wave velocity, elastance of the left ventricle, systemic peripheral compliance, and the duration of ventricular relaxation, each tailored to varying age groups.⁴⁵

In a related study, researchers presented a closed-loop mechanistic model that included an electrical circuit, representing blood pressures as voltages, blood volumes as charges, and blood flow rates as currents. It contains components for the heart, valves, arteries, veins, and cerebral circulation. The mechanistic model incorporates activation functions characterizing the ventricular contractions and elastance functions characterizing the systolic and diastolic elastance of ventricles to generate the ballistocardiogram in normal circumstances and in settings of heart failure with reduced ejection fraction.⁴¹

By incorporating mechanistic models that can generate the ballistocardiogram, a digital twin tailored to specific individuals can simulate the continuous dynamic behavior of their cardiovascular system based on the continuous measurement of the ballistocardiogram. In addition, health care providers can interpret the ballistocardiography for early detection of heart conditions, risk assessment, and predictive analytics. Furthermore, these mechanistic models can help simulate the impact of medications, lifestyle changes, or medical procedures on the ballistocardiogram, allowing for more personalized and effective treatment planning. Incorporating the ballistocardiogram generation into a digital twin opens up the unprecedented possibility to simulate various heart conditions and examine their influence on the ballistocardiogram. This capability could be used to synthesize the ballistocardiogram reflecting diverse cardiac states, enabling an in-depth study of the relationship between heart physiology and the ballistocardiogram. Enhancing the complexity of the mechanistic models by incorporating additional details (including factors like nonconstant velocity values across various arteries) can yield a more precisely generated ballistocardiogram. Clinical application of these mechanistic models is currently limited due to a lack of personalization. The ballistocardiogram can differ between individuals due to differences in body composition (height, weight, and body surface area), sex, and cardiac health.⁵⁷ Not considering patient variability can result in imprecise evaluations and diagnoses and limit the ballistocardiogram's clinical value. Gathering patient-specific information and adjusting the parameters influencing the ballistocardiogram such as heart rate and stroke volume allows for the development of personalized ballistocardiogram generators. The

development of mechanistic models based on data from wearable devices offers the potential for continuous updating of the digital twin with the ballistocardiogram, ensuring more accurate health monitoring.

Ventricular and Atrial Tachycardia Ablation Planning Using Computational Modeling

Ventricular and atrial tachycardia are life-threatening arrhythmias that often occur after myocardial infarction and may require catheter ablation therapy that is designed to ablate electrical pathways involved in sustaining the arrhythmias. Because identifying areas to ablate is often challenging, the availability of patient-specific computational cardiac system modeling could help identify ablation targets. Recent research presents an approach called OPTIMA for personalized, optimal, and targeted ablation of persistent atrial fibrillation in patients with atrial fibrosis. OPTIMA uses the patient's late gadolinium enhancement MRI scans to construct a 3D atria model. The provided model involves patient-specific atrial geometry and fibrotic and non-fibrotic tissue and doesn't include patient-specific electrophysiological parameters. An image intensity ratio-based approach calculates the late gadolinium enhancement magnetic resonance imaging (LGE-MRI) signal intensity in the atrial wall, differentiating fibrotic regions from normal tissue. This provides detailed patient-specific maps of fibrosis distribution to the computational models. Simulations are run to determine potential sources of atrial arrhythmias by analyzing model responses to rapid pacing from 40 sites in both atria.⁵⁸

Most current modeling approaches rely on MRI data for personalization, which is challenging in patients with ventricular tachycardia with implantable defibrillators. Some researchers have presented a framework for generating a personalized electrophysiological model of the heart using computed tomography (CT)-derived myocardial wall thickness to parameterize the conduction velocity therein. This framework used the Eikonal model, a simplified representation for waveform propagation, to simulate cardiac activation maps with improved performance; see Figure 5. This framework is especially suited for clinical applications by virtue of limited human interference and its ability to translate imaging data into electrophysiological simulations within minutes³⁸ and can be integrated in a cardiac digital twin. However, reliance on periodic CT images for updates limits its application within the digital twin framework.

Patient-Specific Ventricular Electrophysiological Modeling

Sudden cardiac death, often from cardiac arrhythmias, remains an important cause of mortality. Individuals with a reduced left ventricular (LV) ejection fraction are

at much higher risk of arrhythmic sudden cardiac death and implantable cardioverter-defibrillators can reduce risk in high-risk patients. However, LV ejection fraction has low sensitivity and specificity for identifying patients who would benefit from implantable cardioverter-defibrillators.⁵⁹ A novel personalized approach called virtual-heart arrhythmia risk predictor has been used to stratify the risk of sudden cardiac death in patients with prior myocardial infarction; see Figure 5.³³ The virtual-heart arrhythmia risk predictor involves several steps to create a personalized virtual mathematical model of the heart: (1) MRI images are acquired to construct 3D ventricular anatomical maps of scar, gray zone, and healthy tissue; (2) highly detailed finite element meshes with assigned fiber orientations are generated automatically using an octree-based approach; (3) region-specific electrophysiological properties are then assigned to the elements in the mechanistic model, designating scar tissue as nonconductive and distinguishing action potential dynamics and conductivities in healthy tissue and the border zone; (4) subsequently, a pacing simulation is executed at various ventricular locations to evaluate the potential for arrhythmias; and (5) the response of the mechanistic model to pacing is analyzed to predict the patient's arrhythmia risk. The integration of such an electrophysiological model into a digital twin enables continuous simulation of a patient's cardiac electrical activity. This simulation serves to assess the risk of arrhythmias following a myocardial infarction by analyzing responses across various pacing locations. However, the existing gap in this literature lies in the challenge of providing continuous updates to the ventricular parameters in the mechanistic model. Establishing such an update mechanism would guarantee continual digital twin feedback regarding the patient's cardiac electrophysiological health.

In another study,⁶⁰ multiscale mechanistic models were personalized for a large cohort of patients at different stages of systolic heart failure, ranging from New York Heart Association Class I to IV. Incorporating cardiac anatomy, electrophysiology, biomechanics, and hemodynamics necessitated a large amount of personal data including pseudonymized imaging, 12-lead ECG, and pressure measurements obtained from catheterization. To demonstrate the potential of their mechanistic model, the authors calculated estimates of parameters not available from clinical phenotyping, such as electrical conductivity and myocardial stiffness. These metrics correlated with biomarkers of heart failure such as NT-proBNP (N-terminal pro-B-type natriuretic peptide) and the Seattle Heart Failure Score. In a similar study, a semi-automated pipeline was presented that integrated many clinical and nonclinical measurements, including CT imaging, echocardiography, cardiac catheterization, methoxyisobutyl isonitrile-single-photon emission CT, and diffusion tensor-MRI to develop multiscale

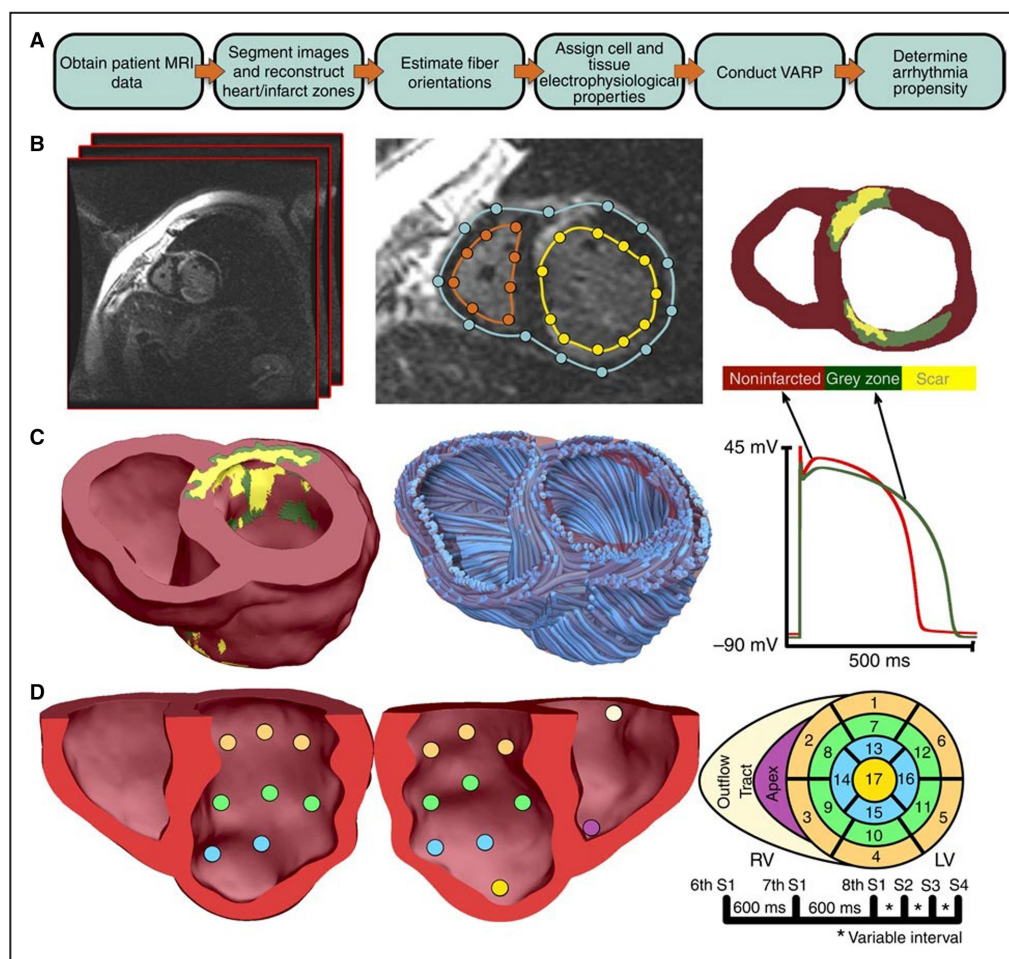


Figure 5. Electrophysiological modeling of the ventricles.

A, Flow chart describing VARP protocol. **B** and **C**, Contrast-enhanced MRI image stacks used for high-resolution ventricular structure modeling with fiber orientation estimations. **D**, VARP pacing sites on the RV and LV endocardial surface, colored following the American Heart Association nomenclature. Reproduced from Arevalo et al.³³ under the terms and conditions of the Creative Commons Attribution (CC-BY) License (<https://creativecommons.org/licenses/by/4.0/>). LV indicates left ventricle; MRI, magnetic resonance imaging; RT, right ventricle; and VARP, virtual-heart arrhythmia risk predictor.

patient-specific mathematical models for 5 patients with LV dysfunction.²⁹ In contrast to the previous study, the endocardial and epicardial contours of each patient's heart were manually segmented from the CT images. In the next steps, several hemodynamic parameters of the circulation submodel were estimated by using the measured cardiac output, mean arterial pressure, and aortic and mitral valve dimensions. In another similar but more limited effort, the unscented transform and reduced-order unscented Kalman filter were used to estimate the contractility of all American Heart Association zones of the left ventricle from cine MRI recordings of healthy volunteers and patients with heart failure.³⁰ The geometry personalization and motion estimation from the image data were performed with a semiautomatic segmentation technique and an automatic registration algorithm. Compared with

the models discussed in the Computational Modeling Combining Electrophysiology and Biomechanics subsection of [Review of Cardiovascular System Modeling Techniques](#), the personalized models in this section more fully embody the concept of a digital twin. They not only replicate the structure and behavior of the heart but also offer a personalized insight able to inform decisions and provide actionable information, thereby guiding health care delivery. However, they still lack the continuous flow of information from the human health system to the computational model and are more of a digital snapshot. Although the current realization of such personalized mechanistic models is possible only with extensive data and expert clinicians, they can still be of great value in applications of digital twins such as therapy planning, surgical simulations, and implantable medical devices.³⁰

Modular Multiscale Mathematical Models for Heart Failure

In an effort to simulate heart failure, researchers have developed a mechanistic model to investigate the impact of LV geometry, afterload, and muscle contractility on myocardial strain in heart failure with preserved ejection fraction.⁴⁴ This multiscale mechanistic model integrates a 3D finite element model of the left ventricle with a lumped-parameter electrical circuit analogy to represent the systemic circulation including the left atrium, arteries, and veins. It provides physiological loading conditions to the finite element model of the ventricle and enables simulation of the hemodynamics and ventricular mechanics such as global longitudinal and circumferential strain. The authors hypothesized that the reduction in LV strain associated with heart failure with preserved ejection fraction can be attributed to either an increase in afterload or a decrease in myocardial contractility. The importance of the mechanistic model is its ability to integrate multiple physiological parameters and investigate their impact on myocardial strain. Better personalization for assessing the impact of fibrosis or inflammation on myocardial strain can make this mechanistic model more relevant to digital twin applications. In another study, novel methods were used to perform coupling finite element models of cardiac mechanics to closed-loop lumped-parameter mathematical models of the circulation.²⁵ This method was used to set up a modular mathematical model of cardiovascular system, which includes a finite element model of the ventricles, a lumped-parameter mechanistic model of circulation, and an algorithm to ensure the continuity of blood flow at the interface between the finite element and lumped-parameter mechanistic models. The mathematical model incorporated 3D left and right ventricular geometry with 3D myofiber angle distribution, time-varying elastance of atria, and Windkessel models of systemic and pulmonary circulations. Most mathematical models were not specifically designed for digital twin applications, and their parameters were either tuned manually or for specific cohorts; however, their breadth in complexity level and their ability to replicate the experimental results in healthy and pathological heart conditions are promising. The idea of a modular, multiscale mathematical model can be extended to enable clinicians to tailor the complexity of specific subsections of a patient's digital twin based on their overall health condition and specific symptoms. This customization aims to achieve the ideal simulation time and physiological granularity.

Three-Dimensional Modeling of an Arterial Aneurysm

The simulation of blood flow dynamics affecting the tissue walls of a particular artery holds significant importance, as it is relevant to aneurysm formation and

rupture.⁶¹ A novel approach to reconstructing carotid artery cerebral aneurysms using personalized 3D computational fluid dynamic modeling of CT angiographic images investigated the effects of pulsatile blood flow motion on the arterial wall and to incorporate mechanical properties for the identification of material failure rate that relates to rupture of a carotid artery aneurysm. Flow simulations, performed under variable wall shear stress and pressure conditions, helped identify regions of high shear stress and pressure that may be prone to rupture.²⁶ For digital twin applications the aforementioned flow simulations can be performed using a single personalized 3D model of a patient's arterial wall and may be updated in a continuous manner using routine medical imaging practices.

Mechanistic and Machine Learning Modeling for Blood Pressure and Pulse Volume Waveforms

Analysis of the arterial pulses (eg, arterial blood pressure waveforms) has the potential to elucidate cardiovascular health and disease. Abnormalities in the arteries can distort the shape of the arterial blood pressure waveforms by altering the characteristics associated with blood pressure wave propagation and reflection.⁶²

Physics-based mechanistic models of arterial hemodynamics can be broadly classified into 2 categories: low-dimensional (0-dimensional, 1D, and tube-load) versus high-dimensional mechanistic models (2-dimensional and 3D). Zero-dimensional models, also known as lumped parameter models, offer a simplified overview of global arterial properties. An illustrative example is the Windkessel model, proposed by Otto Frank in 1899,⁶³ which provides a simplified representation of the arterial system, featuring two primary components: compliance and resistance. On the other hand, 1D models and tube-load models belong to the category of distributed parameter models. These mechanistic models provide a highly accurate representation of wave propagation in arterial systems but necessitate more computational resources. High-dimensional models, encompassing 2-dimensional and 3D models, offer insights into local blood flow dynamics.⁶⁴

Here, large arteries are modeled as (elastic or viscoelastic) tubes transmitting the blood pressure waves, while small peripheral arteries including arterioles are modeled as lumped-parameter loads (eg, Windkessel) to estimate central aortic blood pressure waveforms.⁶⁵ Noting that (1) the tube-load model relates blood pressure waveforms at various sites in the arteries, and (2) arterial blood pressure waveforms are not amenable to wearable-based convenient and noninvasive sensing, emerging work in this field investigates the

extension of the tube-load model to enable the analysis of noninvasive arterial pulses resulting from the arterial blood pressure waveforms (eg, pulse volume recording, ballistocardiography, and photoplethysmography). In one work, researchers developed mechanistic models to relate arterial blood pressure waveforms to pulse volume recording waveforms.³⁹ This presents a framework to estimate central aortic blood pressure within a digital twin, using pulse waveforms measured in the upper arm, wrist, finger, or ankle.

Machine learning-based methods to infer cardiac parameters from arterial pulse waveforms are increasingly being reported.⁶⁶ Studies have reported accurate inference of LV parameters (including the ventricular elastance, the maximum baroreceptor gain, and steepness of the pressure-volume curve) from systemic and pulmonary arterial blood pressure waveforms or brachial artery blood pressure based on a multilayer neural network based on in silico data generated by a lumped-parameter or a 1D mechanistic model. Likewise, the use of in silico clinical trials, which can computationally model the effectiveness of medical approaches, can be used to further simulate these cardiac mechanisms.⁶⁷

Personalized Lumped-Parameter Models for Cardiomyopathies

For a few cardiac conditions, efforts have been made to enable personalization of a lumped parameter model to the pathophysiological condition of a patient of interest. Reviewing these efforts as a rudimentary organ-level digital twin can provide insights regarding practical limitations in realizing a whole-body digital twin. In one work,⁴² a conventional electric circuit-analog mechanistic model of the cardiovascular system was augmented with a mechanistic model of time-varying heart chamber geometry to simulate cardiac function at the organ level. The authors in particular simulated clinical indicators of cardiac function such as sphericity index and fractional shortening, which are used in the diagnosis of heart failure. In a follow-up work,⁴⁹ the authors personalized the mechanistic model to 3 clinical dilated cardiomyopathy cases in children, where the upper and lower bounds of 23 ventricular and hemodynamic parameters in the mechanistic model were optimized to fit the cardiac output, mean arterial blood pressure, and ventricular geometry data. In another work,⁴⁷ researchers personalized a lumped-parameter mechanistic model called the CircAdapt Model, to echocardiographic data of 9 arrhythmogenic cardiomyopathy patients using the adaptive multiple importance sampling method. The personalized parameters were 2 global parameters describing the relative systole duration and cardiac output and 18 regional parameters describing the constitutive behavior

of active and passive stress, activation delay, and reference wall area. To account for the uncertainty present in both the measurements and the mechanistic model, the posterior distribution of regional ventricular deformation patterns and the underlying tissue properties were estimated instead of providing only point estimates. The regional deformation characteristics were simulated closely to the measured counterpart with a reasonable uncertainty profile. However, in the absence of a ground truth for the estimated tissue property parameters, only the reproducibility of these estimates was evaluated, taking into account 3 kinds of variability, namely computational, interobserver, and intraobserver. In another work,³⁶ a combination of local identifiability, Bayesian estimation, and maximum a posteriori simplex optimization was used to estimate about 30 parameters and their uncertainty in a lumped-parameter mechanistic model for a cohort of 4 single-ventricle patients. For this purpose, data obtained from catheterization-derived blood pressure tracings and MRI flow tracings were used. By testing different computational frameworks, the authors showed that employing a multilevel Bayesian estimation approach, that is, updating the prior parameter information through submodel analysis, could lead to a reduction in the marginal posterior parameter variance by taking advantage of the compartmental nature of the lumped-parameter mechanistic models. Although the mentioned examples are rudimentary in key elements required to realize the concept of a dynamic and self-updating diffusion tensor, they all point out the diffusion tensor's capability in providing health care professionals with a clear and engineering-enabled assessment of unmeasurable patient parameters with uncertainty profiles.

Interconnected Cardiopulmonary Modeling

In the process of modeling the cardiovascular system, it is not uncommon to incorporate its interactions with other organ systems.⁶⁸ For example, there is clearly a vital relationship between the cardiovascular and pulmonary systems that involve blood oxygenation and gaseous exchange. Researchers have developed an integrated mechanistic model of the cardiopulmonary system in order to simulate the physiological interactions between these systems.³² Specifically, the mechanistic model is able to describe the dynamic behavior of the heart, blood vessels, and lungs in hypoxic and hypercapnic disease states. Mathematical expressions pertaining to the circulation of blood, gas exchange in the lungs, and short-term neural control mechanisms are used to create the mechanistic model, enhancing the realism of the simulated physiological data acquired. Overall, the cardiopulmonary

model contains hundreds of differential and algebraic equations to quantitatively describe the dynamic relationship between the cardiovascular and pulmonary systems. In addition, the outputs of the mechanistic model allow researchers to analyze vital modulation relationships between physiological biometrics such as tidal volume, LV stroke volume, and blood pressure and were validated with real patient data at these conditions.³² In a similar work, other researchers have developed a detailed mechanistic model of the human cardiovascular-respiratory system to recapitulate the settings of congestive heart failure.²⁷ A few parameters of this multicompartiment mechanistic model were adjusted to simulate the hemodynamic and respiratory signatures of LV diastolic dysfunction. Using the mathematical representation of the cardiovascular-respiratory system, this work showed that LV diastolic dysfunction causes heart failure with commonly recognized signs of decreased cardiac output, stroke volume, and mean arterial blood pressure, with widening of the arteriovenous oxygen difference and pulmonary congestion. Moreover, this work pointed out signatures of LV diastolic dysfunction not previously recognized, such as the abnormal tricuspid flow patterns and right ventricular ejection fraction. The integrated relationship presented in these mechanistic models is an essential attribute of the digital twin capabilities allowing for the expansion of the cardiovascular system model to multiple organ systems, continuous health updates affecting multiple systems, and personalized disease progression. Nevertheless, the mechanistic models developed in these studies are limited by their capabilities for personalization to specific individuals. They lack integration with a sensor modality that prevents continuous updates with respect to a physical twin and with sensor integration would provide the patient specific parameters from 2 organ systems, cardiovascular and pulmonary, to be incorporated into the digital twin. The opportunity for the interconnected cardiopulmonary mechanistic model to be integrated with readily available respiratory and cardiovascular sensors would be the first step toward development of a multiorgan system digital twin, where the hierarchical relationship between gas exchanges in the lungs and resulting cardiovascular dynamics is established.

Integrated Modeling of Cardiorenal System

The interactions between the cardiovascular and renal systems are crucial to maintaining fluid volume and cardiac output.⁶⁹ For a better understanding of conditions that affect the cardiorenal systems, such as heart failure and diabetes, researchers have developed a mechanistic model that can simulate the interactions between the 2.⁴³ A model tasked with simulating heart

failure with reduced ejection fraction and its response to administration of SGLT2 (sodium-glucose cotransporter 2) inhibitors estimated both the preload and both the blood and interstitial fluid volumes. Results indicated that SGLT2 inhibitors successfully decreased the cardiac preload and the interstitial fluid volume without decreasing total blood volume, demonstrating the potential use for this integrated mechanistic model.⁴³ The significance of this particular mechanistic model lies in its ability to replicate the relationship between the cardiovascular and renal systems, which is an important aspect for a digital twin to reproduce the interactions between multiple physiological systems, thereby ultimately enhancing the overall realism of simulating the physical twin and expanding its predictive power. The mechanistic model is able to personalize the 2-baseline hemodynamic and renal characteristics of a patient but does not have continuous measurement capabilities, limiting its current integration within a digital twin.

OPPORTUNITIES FOR A CARDIOVASCULAR DIGITAL TWIN

A digital twin for cardiovascular health leverages observations and information on the unique physiology of the individual to calibrate and update the virtual representation established through mechanistic models and simulations that are most often assisted by AI. Mechanistic modeling of physiological systems has seen tremendous progress over the past half a century, some of which are highlighted in this paper. However, transforming these mechanistic models into clinical practice has not been fully realized yet, due to unmet gaps pertaining to the challenges associated with personalization and contextualization of these mechanistic models. Table 3 provides an overview of identified challenges that need to be addressed for successful implementation of a cardiovascular digital twin and recommendations by the authors to overcome these challenges.

Personalization requires the inference of unknown parameters in the mechanistic model. However, mechanistic models of cardiovascular physiology are typically vastly complex with many tunable parameters that need personalization, which may not be feasible.⁷⁰ Therefore, the majority of modeling efforts has been directed toward constructing mechanistic models that are either comprehensive yet could provide only a representation of an “average” individual or are personalized yet could provide only a limited representation of a specific function in the cardiovascular phenomena that has very narrow applications.⁷¹ In addition, it is crucial to recognize that most existing cardiovascular modeling work do not implement continuous updating

Table 3. Challenges, Barriers, and Opportunities

	Existing challenges and barriers need addressing for translation	Disciplines to address existing challenges	Recommendations and opportunities for successful realization of digital twins
1	Personalization with mathematical modeling that often provide “average” population representations.	Data and computational physiology sciences	Mathematical and computational modeling advances and data-driven modeling approaches should be integrated to support flow of personalized information from heterogeneous ambulatory data
2	Continuous update of mathematical models with integration of sensor data.	Microelectronics & IoT engineering	Development of methods prioritizing interoperability and streamlined integration with an emphasis on verification, validation, and uncertainty quantification is necessary. Example areas include, data/sensor/model fusion, edge computing, continuous learning
3	Artificial intelligence’s requirement of sufficient training data obtained at personalized levels from diverse health states (eg, healthy, and diseased).	Data and clinical sciences	Additional technical work is needed. Example approaches include scientific machine learning models, synthetic data generation approaches, and data augmentation techniques
4	Integration of heterogeneous data types, eg, genetic, clinical, wearable sensor data.	Data science, IoT engineering, and clinical science	Data and sensor harmonization, assembly, and aggregation methods, as well as data categorization and augmentation methodologies are needed
5	Integration of mathematical models built for cells, tissues, organs, and organ systems based on their interactions.	Cellular biology and computational physiology sciences	Cross-disciplinary collaborations should be supported. Required complexities for digital twin applications need proper identification and standardization
6	Formulating a prioritized development roadmap for a digital twin’s integration into the clinical workflow.	Clinical administration and regulatory affairs	Cross-disciplinary collaboration fostering harmonization, standardization, and infrastructure development for digital twins through involving government entities, academia, industry, and health care systems should be encouraged
7	Establishing FDA’s regulatory framework for an “evolving” digital twin	Regulatory affairs	Regulatory science driven by FDA in collaboration with all stakeholders should develop strategies for de-risking and validation of “dynamic” health care systems, while supporting a translational ecosystem fostering innovation
8	Facilitating trust and physician adoption with a digital twin	Data and computational physiology sciences, clinical administration	Verification, validation, and uncertainty quantification techniques should be implemented, while effectively and transparently communicated to all stakeholders
9	Addressing ethics and privacy concerns raised by the highly personalized nature of a digital twin.	Cybersecurity, data science, and clinical administration	The ethical and privacy considerations should be implemented for all digital twin elements and should made through the life cycle of digital twins
10	Equitable design of a digital twin to provide an unbiased representation of all individuals.	Data and computational physiology sciences	Biases in data, models, and accepted clinical paradigms should be considered in developing and maintaining digital twins

FDA indicates Food and Drug Administration; and IoT, Internet of Thing.

paradigm based on new measurements. Instead, the majority of the existing approaches often provide a “digital shadow” or “digital snapshot” of the physical counterpart. A digital twin for cardiovascular health should integrate continuous data collected from individuals and AI-driven data analysis to construct personalized mechanistic models that encompass underlying cardiovascular physiology and mechanics. This will also enable us to continuously update the mechanistic models based on individuals’ varying health status.

AI plays an instrumental role in the development of a digital twin and in deriving actionable information with the assistance of mechanistic models embedded in the digital twin. AI can also play as a surrogate for parts of a digital twin that are otherwise challenging to construct and personalize. Given the formidable complexity of the cardiovascular system, mechanistic modeling may fail to represent certain parts of a digital

twin, in particular certain input–output relationships for parts of the biological system with an aspiration of deep personalization. In such scenarios, AI can learn from data to approximate the system’s dynamics, while streamlining digital twin construction by automating data assembly, categorization, and flow from physical twin to digital twin. For instance, prior work leverages deep learning for standardizing medical image segmentation and automation of aortic stenosis modeling, making the data suitable for large-scale computational modeling.⁷² AI can also assist in extracting actionable information from sensor measurements that in most cases would only provide a proxy for the underlying physiological phenomena.⁷³ A prior work performed a prospective study involving over 400 000 UK Biobank patients, where they built ensembled machine learning algorithms to predict cardiovascular disorder events in asymptomatic individuals.⁷⁴ They showed their

machine learning modeling framework, leveraging over 450 variables for each participant, outperformed the well-established traditional techniques in performing accurate cardiovascular disorder risk assessments. In the digital twin context, this can become feasible due to the mechanistic models boosting and complementing the measurement and effectively putting measurement into perspective. However, data-driven AI algorithms require a large amount of ground truth data collected from diverse health states (eg, healthy and diseases) for training to accurately represent its real-world counterpart and to avoid overfitting to noise, artifacts, and peculiarities in the data that are not representative of meaningful physiology. In most biomedical applications, access to ground truth data, especially at the individual levels, creates significant burden, challenges, and limitations. Moreover, in some cases, person-specific ground truth data from a disease may not exist (eg, collecting early training data from a stroke event is not feasible). One opportunity to bridge this gap is to guide AI training with the underlying physical laws defined for the system, such as in the case of physics-informed neural networks (PINNs). PINNs use mechanistic equations, initial conditions, and boundary conditions of a physical system to enforce AI estimations to be consistent with the underlying physical laws.⁷⁵ During training, PINNs use AI's inherent auto differentiation function to solve the equations and minimize a residual error between a numerical mismatch between AI's data-driven estimations and physical laws. PINNs show immense success in building robust machine learning models with less data for various physical systems that have well-defined physical laws, for example, explicit partial differential equations.^{75,76} However, their implementation for the cardiovascular system is challenging, because the underlying dynamics are often not explicitly defined and include unknown coefficients with values unique to each individual. A prior work proposed to use Taylor series approximation of the underlying hidden and gradually changing cardiovascular dynamics for the input-output relationships of physiological time series to construct PINNs for time series.⁷⁷ They tested their approach in a case study of cuffless blood pressure estimation using wearable bioimpedance signals and showed that using PINNs with Taylor approximation retains high estimation accuracies while reducing the need for ground truth data by a factor of 15. Another potential direction is generating synthetic data sets to mitigate challenges associated with data limitations and missingness. Recent body of work includes electrophysiological and statistical shape models of atria for ECG simulations,^{78,79} Generative Adversarial Networks and auto-encoders for medical image and time-series data generation,^{80,81} and collective variational inference for generating physiologically plausible and realistic

virtual data on cardiovascular hemodynamics.⁸² Many of these attempts need further validation as such synthetic data are often prone to bias and personalization issues raised from interpatient variability.⁸³

Parameter estimation methods are the key to realize the overarching visions underlying the digital twin concepts by personalizing given mathematical models. Successful parameter estimation hinges upon a number of interconnected factors, including (1) identifiability and sloppiness of the mathematical model; (2) data availability for successful tuning of the mathematical model; and (3) the possibility of overfitting and becoming trapped in local solutions, to list a few. Especially challenging in the context of realizing a digital twin for medical and health care applications is the severely limited access to patient-specific clinical data and the lack of useful information therein to effectively personalize the digital twin. Clinical measurements are (1) strictly limited to those essentially relevant and necessary to treatments as well as (2) passively obtained without any excitation to elicit desired changes in clinical responses. The measurements may be noisy, which poses additional challenges. These challenges have been demonstrated in a few examples in prior work.⁸⁴ Given the potential use of a digital twin in high-risk clinical decision making, establishing trust in the virtual representation is critical, requiring proper embedding of verification, validation, and uncertainty quantification processes.¹ The body of literature covering uncertainty quantification has been presented in several reviews.^{85,86} Although recent work started to incorporate parametric sensitivity and identifiability analysis to obtain robust inverse problem solutions^{36,87,88} as well as Bayesian inference methods or polynomial chaos expansion to quantify parametric and modeling uncertainty,^{47,89} existing work still tends to use rudimentary nonlinear optimization techniques^{49,60} and custom-designed empiric techniques with no strict convergence proof to personalize mathematical models, without legitimate accounting for the possibility of overfitting.⁹⁰ To address this challenge, a large amount of effort has been invested to develop effective parameter estimation methods and related support methods to personalize complex mechanistic models to individual patients using highly scarce clinical data pertaining to the patients. Representative examples (which are by no means intended to be exhaustive) include global optimization via metaheuristic combined with regularization,⁹¹ automated regularized global optimization with sampling-based parameter bounding,⁹² evolutionary algorithms, simulation-based maximum likelihood estimation,⁹³ Markov chain Monte Carlo techniques,⁹⁴ computationally efficient meta model approximation of mixed-effects models,⁹⁵ collective hierarchical variational inference,⁸² Bayesian method with the ability to account for various types of constraints (via rejection

sampling followed by classical parametric inference to approximate the prior probability density function of generated samples),⁹⁶ cooperative metaheuristics method,⁹⁷ quantification of parametric information gains available in measurements,⁸⁸ estimation of CIs using constrained optimization,⁹⁸ detection of unidentifiable parameters in complex mechanistic models,⁹⁹ advanced sampling techniques, unscented transform and unscented Kalman filtering,^{30,35} and machine learning-based methods.¹⁰⁰

The integration of novel sensor measurements and data at different levels plays a pivotal role in the construction and personalization of a digital twin. Yet, this integration presents substantial challenges due to significant differences in data characteristics (eg, genomic data involving discrete genetic markers versus wearable sensor data in the form of continuous time series), complexity, and scale (eg, historical data versus real-time data).¹⁰¹ Hence, data integration becomes nearly impossible without the assistance of statistical models. AI and mechanistic modeling can harmonize these heterogeneous data sources by extracting meaningful patterns, correlations, and insights, even in cases where the quality of the sensor measurements degrade in free-living settings due to common reasons, including sensor level noise and data missingness. AI and mechanistic models can also drive the future of sensor development, inspiring innovation toward creating sensors tailored to the requirements of a digital twin, reversing the traditional approach of developing AI for existing sensor measurements. This shift promises to unlock new dimensions of data and insights for a digital twin, augmenting our understanding of the limitations on what current sensors can offer for precision health delivery. The hierarchical view of a digital twin and composability of personalized mathematical models for cells, tissues, organs, organ systems, their interaction, and eventually a full body digital twin are among our aspirations. Mathematical models that are accurate, valid, robust, and truly represent an individual and her/his organ systems and are constructed bottom up may enable computational medicine to derive extraction of actionable information in an unprecedented way.⁶⁸ This unprecedented opportunity may become possible due to our deeper understanding of the cardiovascular system at all levels abstracted by mechanistic models and their hierarchical relationship complemented by advancements in sensing, phenotyping, and leveraging multi-omics information, as well as AI and the computational medicine. The existing body of work reviewed does not inherently offer the actionable information shared in [Figure 1](#). Future efforts should concentrate on engaging with end users, including care providers and patients, to ascertain the necessary actionable information. Subsequently, a digital twin should be integrated into the clinical system to deliver the required information.

Verification and validation of the mathematical models through careful experimental design would be among some of the most important directions as described earlier. A significant amount of multidisciplinary effort over the years has gone into establishing standards for the verification process.¹⁰² In 2018, the American Society of Mechanical Engineers published the V&V40 Standard to set a framework for assessing adequate techniques to properly establish credibility of a computational model based on factors including code and calculation verification, as well as the model and data validities.¹⁰³ The credibility framework was also implemented into medical devices by the US Food and Drug Administration (FDA) as a guidance, with the most recent update published in 2024.¹⁰⁴ Prior demonstrations of implementing verification, validation, and uncertainty quantification process to electrophysiological modeling in lieu of the V&V40 Standards exist.^{102,105} Yet, given the diverse nature of data and sensor types, statistical and mechanistic models, and underlying assumptions in building a digital twin, validation proves to be difficult.¹ To address this, the digital twin framework would need to be fully integrated by multidisciplinary domain experts, including cardiologists, engineers who work on novel sensors, computer scientists and engineers who develop domain-specific AI algorithms, and experts in mechanistic modeling from biologists and physiologists to physician scientists. Sensor development may not be decoupled from clinical translation and mathematical model refinement. In fact, the real opportunity is to develop sensors to answer the demand of clinical applications and AI, contrary to the current practice that aims at developing AI for existing sensors. Another opportunity is to develop ambulatory wearable sensors that can work reliably in free-living settings.¹⁰⁶ The end-to-end and integrated nature of experimental validation requires larger teams and careful design of the study, power analysis, and mitigation strategies. Digital twin platforms for cardiology may unlock the real opportunities for improved care and offer actionable information only when they are constructed rigorously to provide a comprehensive view of the patient or participant. This may not be fully realized through discrete and isolated sets of experimental validations. Verification, validation, and uncertainty quantification will also serve as a catalyst for the adoption of a digital twin in clinical settings and instill trust among physicians. A near-term opportunity involves developing a straightforward cardiovascular digital twin, such as leveraging AI and mechanistic models to predict future cardiovascular parameters from current and past observations, to foster trust, while gradually progressing toward a more intricate and hierarchical cardiovascular digital twin.

The regulatory implications and the FDA's views on a digital twin are multifaceted and require careful

consideration from both effectiveness and safety standpoints. Considering the deeply personalized nature of a digital twin, where mathematical models adapt to individuals and continuously evolve over time, defining a new regulatory framework becomes of paramount importance. Current regulatory standards for medical technologies cannot address the requirements of the digital twin framework given proposed technology undergoing regulatory approvals require to be fully completed and unalterable, which is incongruent with the dynamic nature of a digital twin.¹⁰⁷ The FDA may view a digital twin as an adaptive technology learning from the users on a second-by-second basis, while considering that (1) a digital twin initially may not have access to all data for adapting to a specific individual, (2) access to more data is dependent on continuous use of the digital twins by those individuals, and (3) a digital twin would evolve toward gaining additional effectiveness and robustness based on new personalized data. A digital twin may not offer its optimal performance at the onset of deployment. Hence, confidence in estimations and extracted information may need to be quantified and presented to the users and physicians. Emphasis should be placed on real-world performance and the ability to work with adverse events, as a digital twin is expected to monitor conditions like stroke and heart failure. Although the complexities of current FDA regulations may instigate concerns among certain digital health entities on pursuing FDA approval, it is essential to strike a balance that fosters innovation while ensuring safety and efficacy, potentially necessitating the development of new regulatory science in collaboration with all stakeholders.

Digital twin technology is inherently highly personalized and provides important insights about individuals and patients. Although this unprecedented feature may be used to significantly affect the patients and their health, ethical and privacy considerations must be taken into account.¹⁰⁸ The unauthorized or unconsented use of such platforms may instigate significant concerns. This requires research and investigation that ensure access to only suitable information, authorized by the participants and patients, and protected through laws and regulations.

Access to equitable health care is among the most fundamental rights and privileges in every society and community. The design of a digital twin, given their integrated and end-to-end nature, must begin with the objective of equity and access for all at the onset of technological development. Given their deep personalized nature, people from all communities, sexes, and ethnicities must be represented in the development of a digital twin. Socio-economic preferences as they pertain to sensing and compliance, cultural implications, and the cost of the technology must be incorporated into study from the early stage.¹⁰⁹ Continuous review

of pitfalls and mitigation strategies, given the scope of the design and development of a digital twin, will need to be integrated into the study. A digital twin for cardiovascular health is not only expected to represent a complex physiological system but also sophisticated human behavior, expectations, and acceptance, which are equally important and pivotal for its successful translation. All these requirements must come together and be deliberately addressed for this magnificent and impactful technology to drive the future of health and medicine in an unprecedented way.

Realizing the full digital twin vision motivated in this paper, building on research such as the examples highlighted herein, will require a significant, sustained effort. It is thus important to create a development roadmap to prioritize research and integrate capabilities into a digital twin architecture that can provide expanding capability over time. The roadmap should identify quick wins and off-ramps to commercialization and prioritize efforts that provide the greatest benefit to health care with the least investment, while at the same time, investing in the long-term research to realize the full vision. The US government and associated National Laboratories have demonstrated a capability to perform this type of systems analysis and architecture definition in other domains, teaming with health care providers and researchers, industry, and academia. In addition, it is worthwhile to mention the cross-disciplinary collaboration with experts, organizations and institutions, National Academies of Sciences, Engineering, and Medicine, EDITH, American Society of Mechanical Engineers, and Auckland Bioengineering Institute (pioneers of the Physiome Project) for their pivotal roles in providing insights, frameworks, and recommendations regarding harmonization, standardization and infrastructure for the digital twin framework.^{1,12,103} With the “4-legged stool” of government, health care system, industry, and academia working together, we foresee an opportunity to transform health care with digital twin technology.

ARTICLE INFORMATION

Affiliations

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Disclosures

Roosbeh Jafari is the founder of SpectroBeat, LLC. Omer T. Inan is co-founder and board member for Cardiosense, Inc. and a scientific advisor for Physiowave, Inc. The remaining authors have no disclosures to report.

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